

## NBA opening-spread walk-forward — model and strategy review

**Strategy:** Directional value-taking against the opening point spread in NBA regular-season games. A market-blind margin model prices every listed game; its disagreement with the opening spread is then spent, at about a third of face value, as a correction to the opening spread —  $m_{\text{final}} = \text{open\_margin} + f(x)$ , with  $f$  a ridge shrunk hard toward zero and fitted walk-forward. A single side is taken where the corrected margin still disagrees with the line by more than a walk-forward selected threshold, at the best number available across books. Every position is one side of one game at  $-110$ , entered at the open and held to settlement: no hedge, no offsetting leg, no in-game management, no exit. Flat 1 unit per bet, so nothing compounds. No calendar filter.

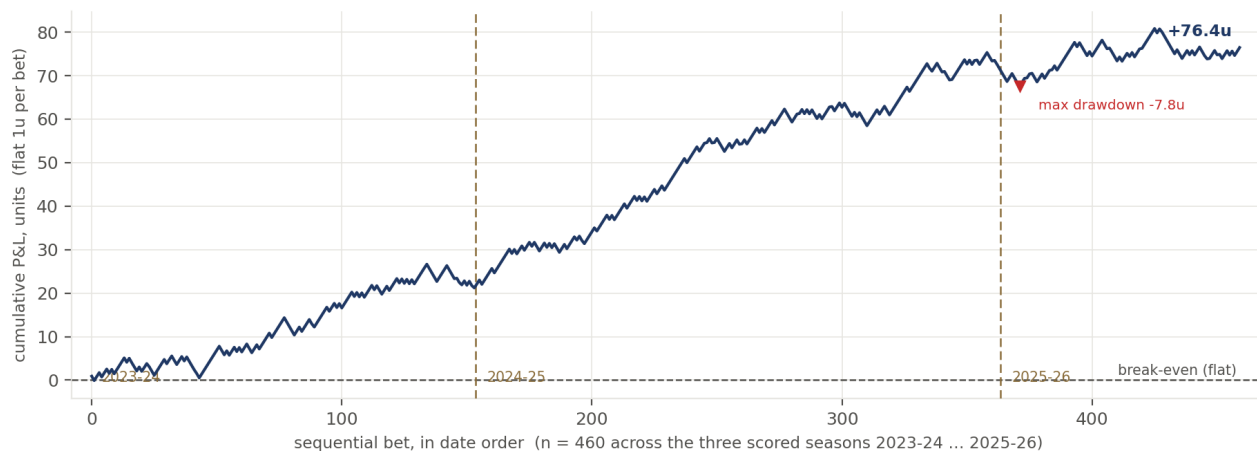
Selection is the strategy. The book is roughly 10% of the slate, so the cover rate on the selected book is not the cover rate on the slate.

**Simulation:** 888 bets across 7 seasons, 2019-20 → 2025-26, on recorded opening spreads. The configuration is selected on seasons 1..k from a pre-declared space, frozen, scored on season k+1, and rolled forward; nothing is refitted on the season being scored, and the selection history reaches back to 2012-13. Earlier seasons are excluded from scoring on principle: the daily injury report begins 2018-12-17 and the availability leg is half the production margin, so before 2019-20 the model cannot run as designed.

### Headline results

| season                                 | books/game at open | bets       | P&L (u)        | ROI            | Sharpe (ann.) |
|----------------------------------------|--------------------|------------|----------------|----------------|---------------|
| 2023-24                                | 7.74               | 154        | +21.20u        | +13.77%        | 1.90          |
| 2024-25                                | 1.00               | 210        | +50.40u        | +24.00%        | 3.81          |
| 2025-26                                | 1.03               | 96         | +4.85u         | +5.05%         | 0.54          |
| <b>2023-26 pooled, best of 9 books</b> | —                  | <b>460</b> | <b>+76.44u</b> | <b>+16.62%</b> | <b>2.25</b>   |
| 2019-26, all seven scored seasons      | —                  | 888        | +80.93u        | +9.11%         | 1.10          |

ROI is P&L per unit staked. Sharpe is annualized by  $\sqrt{\text{sessions per season}}$ , not  $\sqrt{252}$ , which would inflate it 1.8×. Prices are the best of 9 books, but that panel is measured only in 2023-24, so the multi-book figures are modelled for every other season, and best-of-N is the strongest assumption in this document — see the caveats. The last row is the full scored frame and is the honest denominator: the three-season block is the recent end of it, not a separate result.



All 460 bets of the three recent seasons, in date order at the opening spread and  $-110$ ; flat 1u, so the path is a running sum and nothing is re-selected within it. Max drawdown is  $-7.75u$  at bet 372. Across the full seven-season frame the path is  $+80.93u$  on 888 bets, with a  $-23.1u$  drawdown; the first three of those seasons lose money and the last four make  $+83.5u$ .

Pooled over the three recent seasons ROI is  $+16.62\%$  with a season-clustered 95% interval of  $[-9.29\%, +37.83\%]$ ; over all seven it is  $+9.11\%$  with an interval of  $[-2.75\%, +16.08\%]$  and an MDE80 of 12.7pp. **Both intervals contain zero.** In the three-season block 2024-25 alone supplies 66% of the P&L.

The frame is also too short to tune on. Measured, a null taking the best of five randomly chosen game subsets buys  $+2.54$  ROI points on average, and every strategy filter tested lands inside that band. Shortening the window does not escape this: across all contiguous windows the best 3-season window returns  $+16.95\%$  against a  $+9.03\%$  average, and the best 4-season window  $+14.92\%$  against  $+9.33\%$  — so roughly  $+5.6$  to  $+7.9$  of any short-window headline is the selection rather than a regime. The full seven-season frame is the only window whose selection cost is zero by construction.

## The model

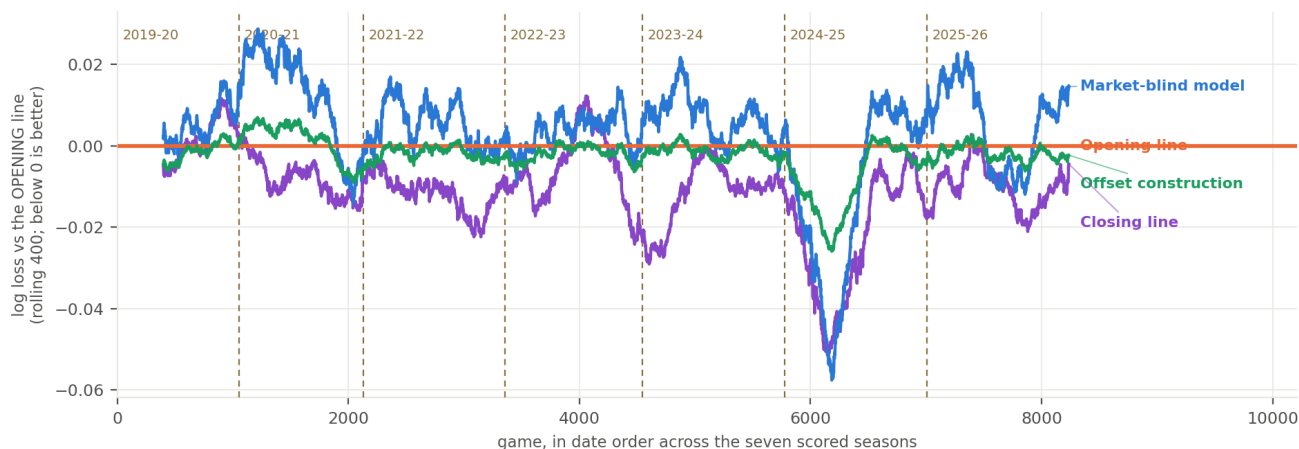
The model outputs a margin, not a probability. A spread is itself a margin forecast, so the sides test involves no devig or probability conversion.

```
margin = 0.5·four_factors + 0.5·availability_composition + schedule_layer + tank_term
P(home win) = sigmoid(margin / 7.2)
```

Two independent estimates of team strength, averaged, plus additive context. The market-blind model never sees market odds, enforced structurally rather than by convention; only the offset layer above it sees the price.

| component                | what it is                                                                                                                                                 | how it is fit                                                                                                                                                                                            |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Four factors             | opponent-adjusted ratings on shooting, turnovers, rebounding and free-throw rate; a decomposition of points per possession rather than a feature shortlist | one L2 ridge solve per factor (ridge=25), then the four adjusted values mapped to points by a fitted linear map. An 8-factor extension tied, so 4 is kept for parsimony                                  |
| Availability composition | $\Sigma$ over available players of DARKO talent $\times$ trailing minutes / 48; the leg that reacts to injuries                                            | each player weighted by $1 - P(\text{out})$ , forecast from as-of-open information only. Requires the daily injury report; this is why the frame starts 2019-20                                          |
| Schedule layer           | home edge, back-to-backs, dead-team flags                                                                                                                  | estimated walk-forward, EB shrinkage $n/(n+600)$ toward a prior, team_home_ridge=200 on per-team home deviations. The only component that has survived strict out-of-sample testing on every split tried |
| Tank term                | late-season effort                                                                                                                                         | exactly zero outside its window                                                                                                                                                                          |
| Offset layer             | the correction to the opening line                                                                                                                         | ridge on three features knowable at the open, shrunk hard toward zero. Fitted edge coefficient 0.33–0.37 in every fold                                                                                   |

## How it compares to the two market prices



Rolling log loss measured against the opening line, which is the zero line, across all seven scored seasons. Levels differ by about 0.01 nats while each series' own rolling path swings 0.30, so a levels chart leaves all four overlapping and illegible; the difference is the readable form. Pooled over 2019-26: opening line 0.60839, closing line 0.59799, offset construction 0.60589, market-blind model 0.61217. This figure keeps the full frame deliberately — forecast accuracy has no reason to be cut to the recent block, and more seasons make the comparison sharper.

**The market-blind model is the only series above the opener.** It does not beat the price it would transact at, and it is beaten decisively by the close. The offset construction sits between the two prices, recovering 24.1% of the open-to-close gap; the blind model recovers –36.4% of it, moving the wrong way.

This is the most useful negative result in the project, and the offset construction does not make sense without it. The blind model is not a discarded attempt — it is the offset layer's dominant input, and the finding that its disagreement must be spent at a third of face value against a market anchor, rather than trusted on its own, is the strategy.

## Bet-time information, and a leak that was found and closed

The availability leg was built on the 5PM injury report and the official pregame inactive list. Both publish *after* the opening line the strategy transacts at, so every open-priced figure produced before this was found gave the model information the bettor did not have.

The exposure was measured rather than assumed. Comparing each team-day's out-set to the previous published report: 81.9% of tonight's absences are already known, 18.1% are new on the day, 19.3% weighted by the absent player's minutes, and 44.3% of team-days carry at least one late scratch. Rebuilding the out-set by carry-forward from the last report strictly before game day widened the model's gap to the market from 12.87% to 17.17% — **the late information was worth 33% of the model's entire deficit.**

The fix does not accept the degraded hard out-set but forecasts it. Each rostered player is now weighted by  $1 - P(\text{out})$ , estimated from as-of-open information only. Players last listed *Questionable* turn out to be out 28.9% of the time; the old hard rule scored them 0.000 and was wrong in both directions. Gated against the hard-rule incumbent with the specification hashed and the detectable effect stated before the endpoint was read: season-clustered delta  $-0.002265$ , 95% interval  $[-0.0041, -0.0004]$  excluding zero, better in 5 of 5 seasons, calibration veto passed. It recovers 52% of the leakage penalty using only what is public at the open.

The correction is also what separates the two architectures. Under honest inputs the market-blind model's capture against the opener fell from  $+0.075$  to  $-0.104$  — it flipped negative — while the offset construction fell only from  $+0.313$  to  $+0.267$ . Because the ridge already spent the model's edge at a third of face value, it was never leaning on the contaminated signal.

## What would change the assessment

- **2026-27, scored prospectively.** The architecture was developed on 2021-26 data, so no retrospective procedure can make it an untouched test. Both arms will run on identical games, with the market-blind arm as the live control.
- **Two books captured at the open, from opening night.** The multi-book logger has never run in-season. If it is down on opening night, that season's open-price record cannot be reconstructed afterwards.
- **More seasons.** Seven is the honest frame and it is not enough. Resolving an effect this size needs roughly 36 seasons at this dispersion.

## Notes and caveats

- Results are **simulated** on recorded opening prices with a walk-forward selection rule. No capital has ever been deployed.
- Multi-book pricing is measured in 2023-24 only (7.74 books/game) and modelled elsewhere; 2024-25 and 2025-26 observe 1.00 and 1.03 books per game at the open.
- **Best-of-N pricing is the single most optimistic assumption here.** Always transacting at the best of nine books means always transacting at whichever book is furthest offside, and those are precisely the numbers that are unavailable in size, moved before the bet lands, or withdrawn from an account that keeps taking them. Applying an outlier charge — no credit for the 8.1% of best-of-N prices sitting more than 1.5 points off the next book — costs about 2.3 ROI points, and even that is a partial model. It does not capture staking limits at soft books, reduced maximums on the best number, or account restriction, which in practice arrive faster than any price effect. Exchange fees and account limits are not modelled at all, and competing flow is absent from the replay.
- Neither arm is the production default; promoting one is a re-certification.
- The full register of 211 decisions, most of them rejections — including four team-name join bugs, an availability leak, two wrong evaluation frames and a confidence interval that briefly claimed false significance — is in docs/DECISIONS.md.